

Graphing quadratic functions

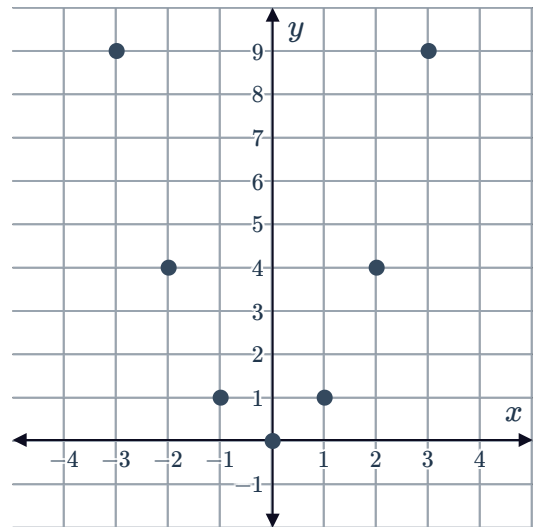


1

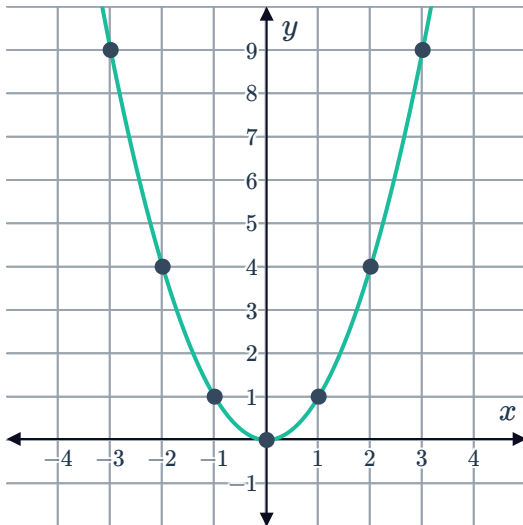
a

x	-3	-2	-1	0	1	2	3
y	9	4	1	0	1	4	9

b



c



d No

e $x = 0$

f 0

g 2

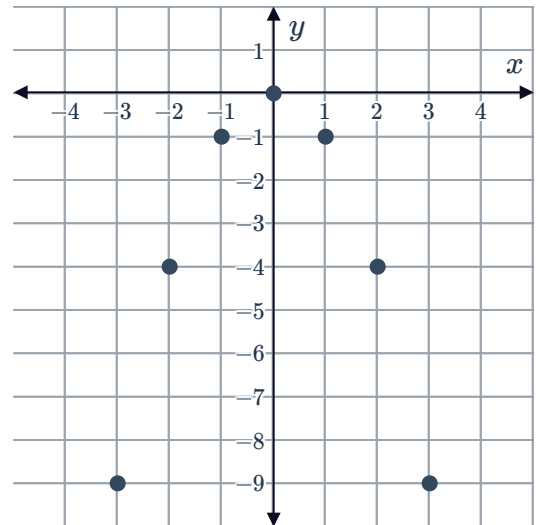
2



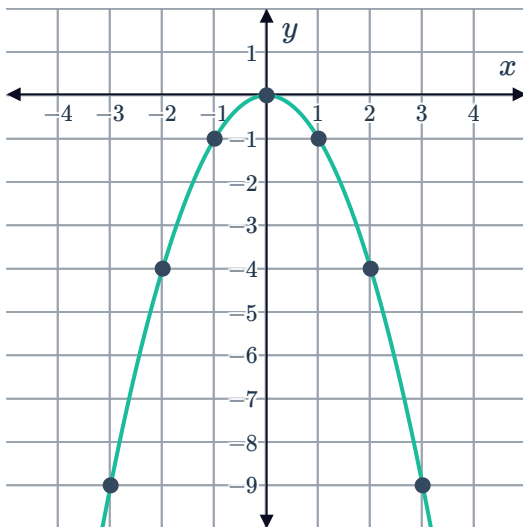
a

x	-3	-2	-1	0	1	2	3
y	-9	-4	-1	0	-1	-4	-9

b



c



d No

e 0

f $x = 0$

3

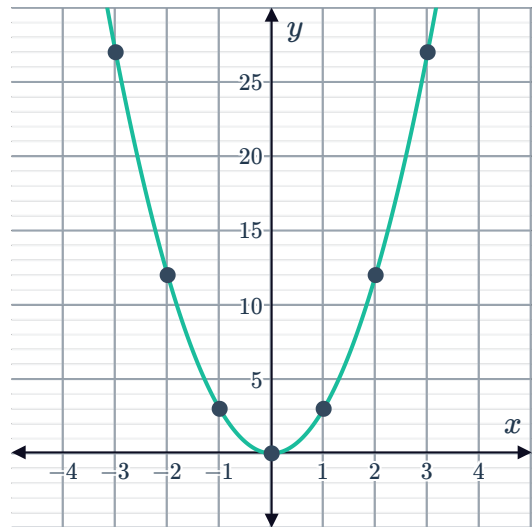
a

i

x	-3	-2	-1	0	1	2	3
y	27	12	3	0	3	12	27



ii

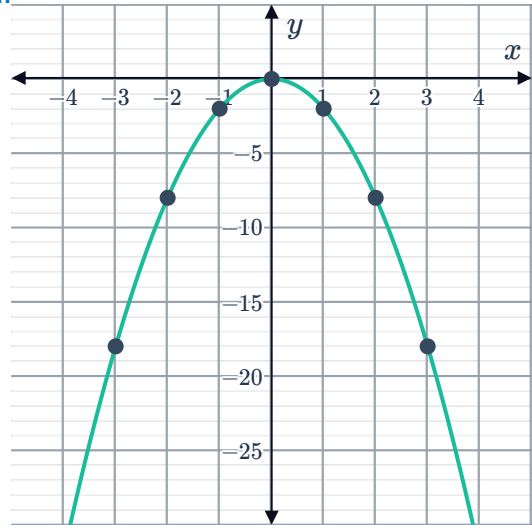


b

i

x	-3	-2	-1	0	1	2	3
y	-18	-8	-2	0	-2	-8	-18

ii



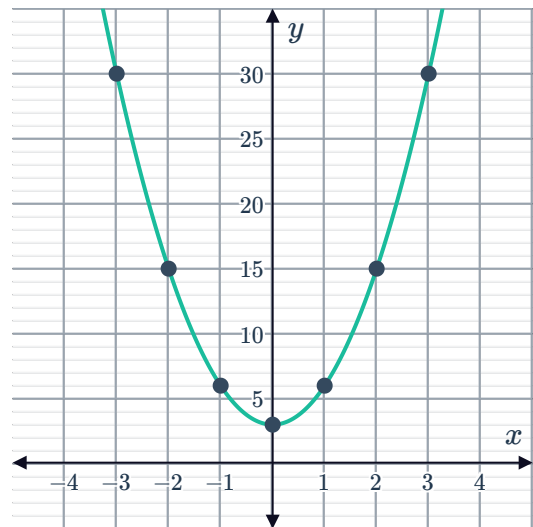
c

i

x	-3	-2	-1	0	1	2	3
y	30	15	6	3	6	15	30



ii

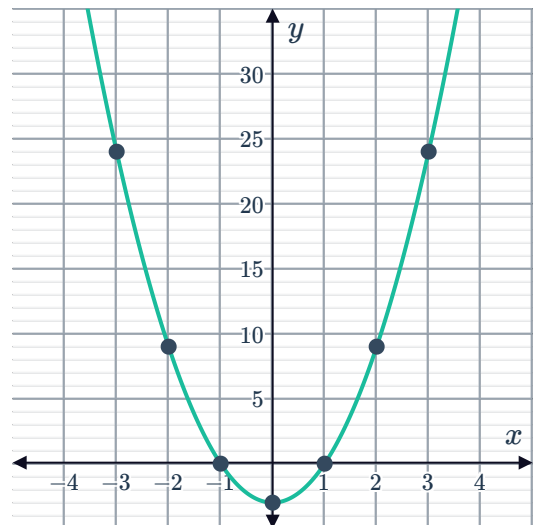


d

i

x	-3	-2	-1	0	1	2	3
y	24	9	0	-3	0	9	24

ii

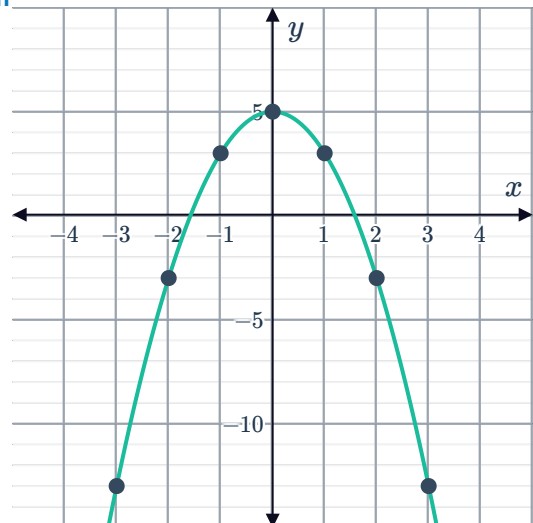


e

i

x	-3	-2	-1	0	1	2	3
y	-13	-3	3	5	3	-3	-13

ii

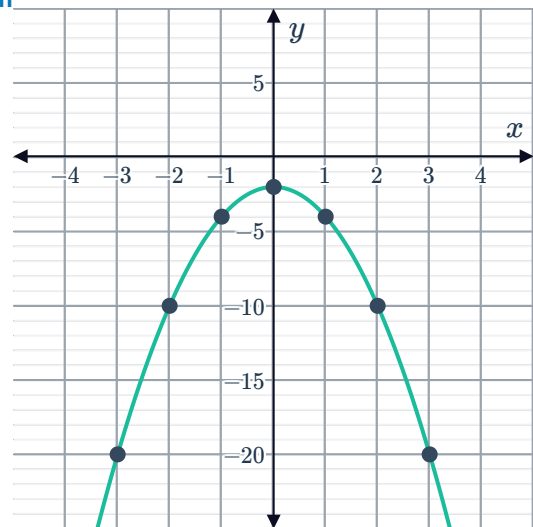


f
i



x	-3	-2	-1	0	1	2	3
y	-20	-10	-4	-2	-4	-10	-20

ii



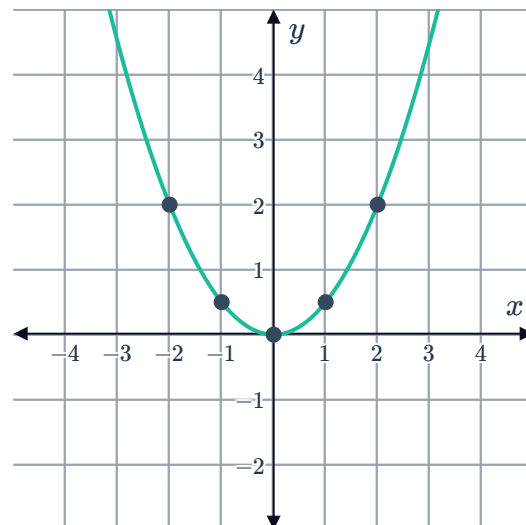
4

a

i

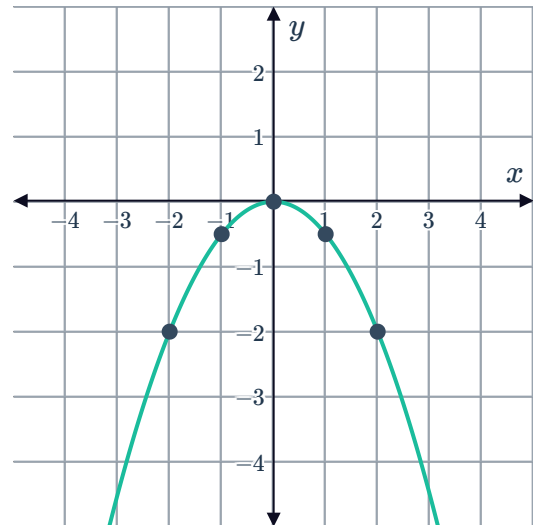
x	-2	-1	0	1	2
y	2	$\frac{1}{2}$	0	$\frac{1}{2}$	2

ii



b
i

x	-2	-1	0	1	2
y	-2	$-\frac{1}{2}$	0	$-\frac{1}{2}$	-2

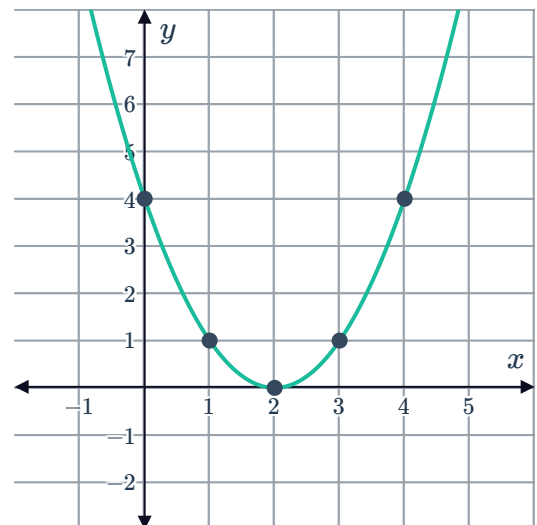


5

a

x	0	1	2	3	4
y	4	1	0	1	4

b



c 0

e (2, 0)

d 2

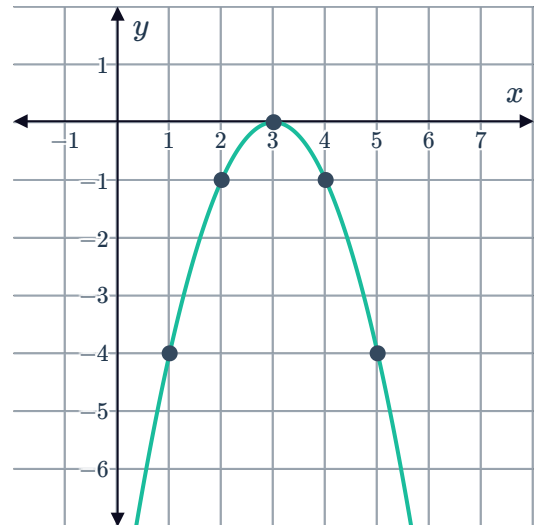
6

a

x	1	2	3	4	5
y	-4	-1	0	-1	-4



b



c 0

e (3,0)

d 3

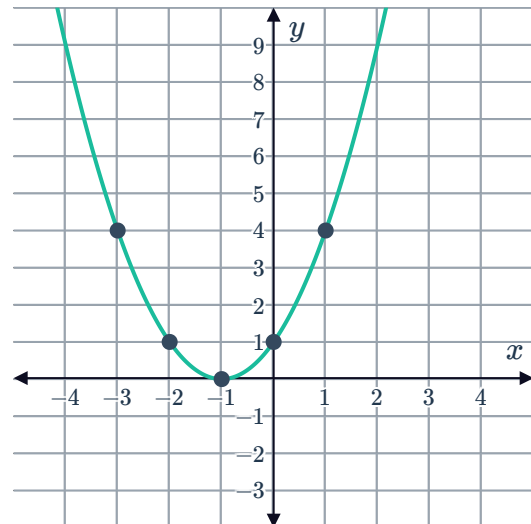
7

a

i

x	-3	-2	-1	0	1
y	4	1	0	1	4

ii



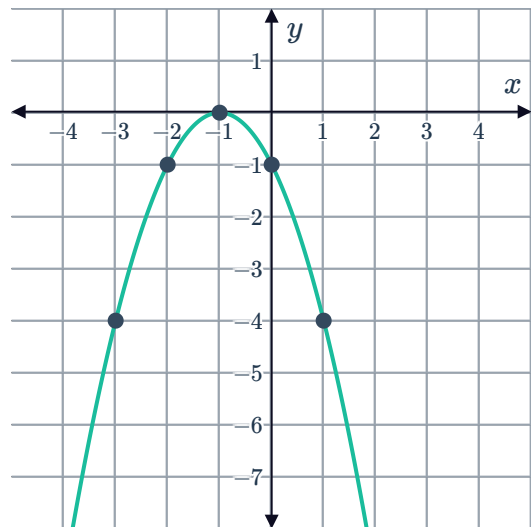
iii 0

v (-1,0)

iv -1

b
i

x	-3	-2	-1	0	1
y	-4	-1	0	-1	-4



iii 0

iv -1

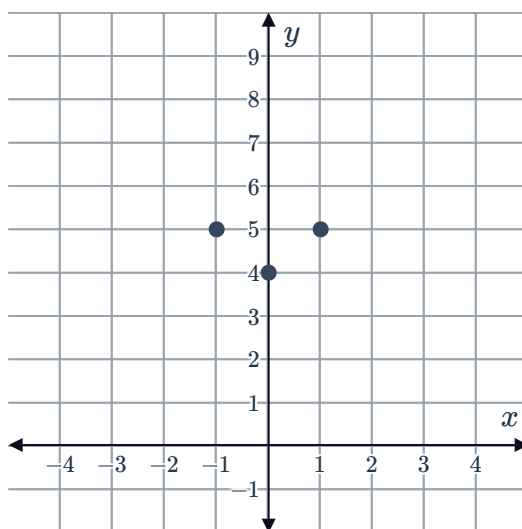
v $(-1, 0)$

8 True

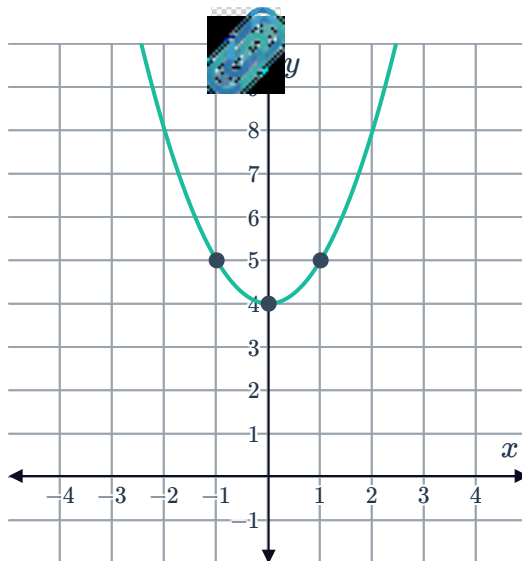
9

a $A(-3, 13), B(-2, 8), C(-1, 5), D(0, 4), E(1, 5), F(2, 8), G(3, 13)$

b



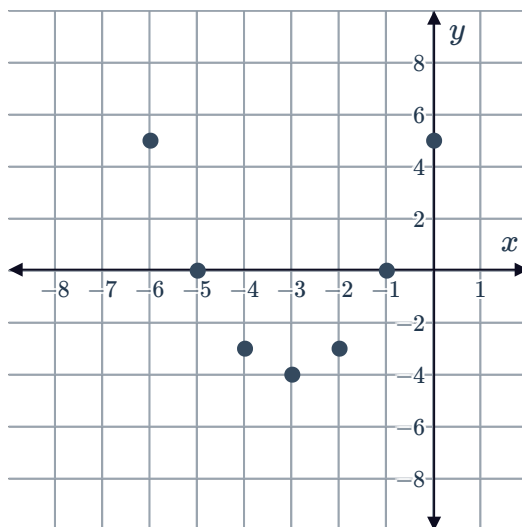
c



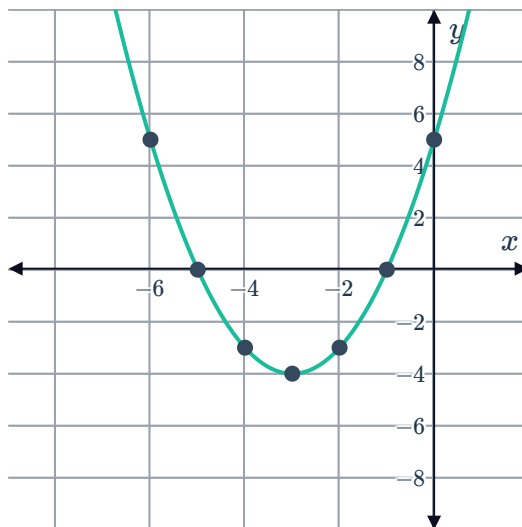
10

a $A(-5, 0)$, $B(-2, -3)$, $C(-6, 5)$, $D(-1, 0)$, $E(0, 5)$, $F(-3, -4)$, $G(-4, -3)$

b



c



11 $y = 0$



12 Maximum value

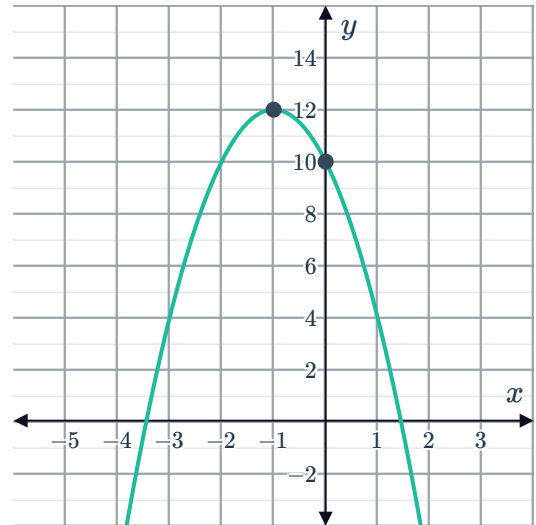
13

a $x = -1$

c $(-1, 12)$

b 12

d

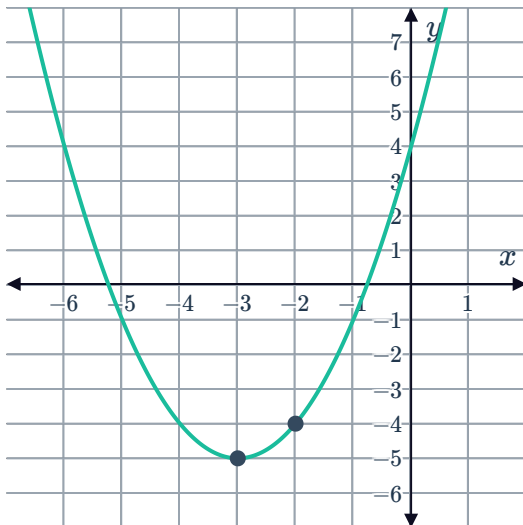


14

a $x = -3$

b -5

c



15

a

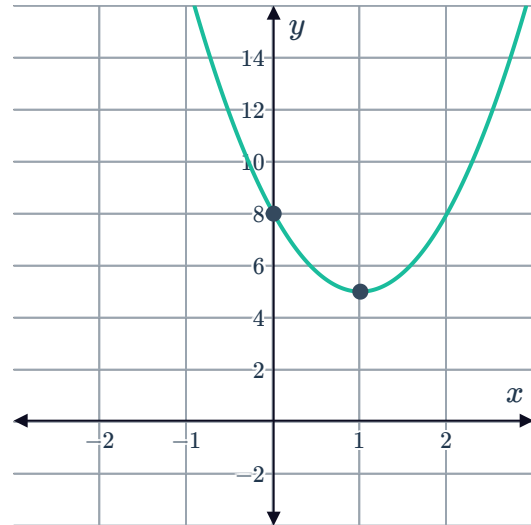
i $x = 1$

iii 8



ii $(1, 5)$

iv



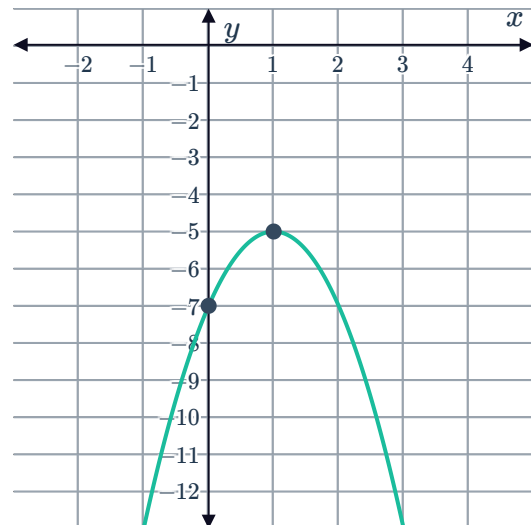
b

i $x = 1$

iii -7

ii $(1, -5)$

iv



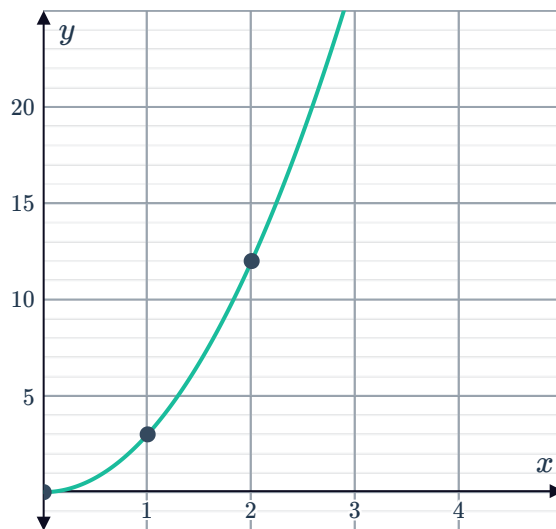
16

a

x (m)	0	1	2	3	4
y (m)	0	3	12	27	48



b



c 0 m

d 0

e (0,0)

17 $g(x)$ has the smallest minimum value which is -16